

Molecules Blooket - Teacher Answer Key

7th Grade Science | 30 questions | 30 seconds each

Learning target

I can make and draw a model of molecules showing the composition before and after a chemical change as well as what happens when temperature changes.

Set design: 10 recall, 10 concept, and 10 application/counting questions. Temperature questions explicitly distinguish particle-motion changes from chemical composition changes. Counting questions stay at a grade-appropriate conservation-of-atoms level rather than introducing stoichiometry.

Answer reference

#	Question	Correct answer and rationale
1	Recall Which statement best describes a molecule?	2. Two or more atoms chemically bonded together A molecule is made of two or more atoms joined by chemical bonds. The atoms may be the same element or different elements.
2	Recall Which is the best example of a molecule?	2. Two oxygen atoms bonded as O₂ O ₂ contains two oxygen atoms chemically bonded together, so it is a molecule.
3	Recall What happens to atoms during a chemical change?	3. They rearrange to form new substances In a chemical change, atoms are rearranged into new combinations. The atoms themselves are conserved.
4	Recall A correct before-and-after particle model of a chemical reaction should show what?	1. The same number of each type of atom before and after Chemical reactions rearrange atoms, so the number of each type of atom must be the same before and after in a closed system.
5	Recall If a sample is heated and no chemical reaction occurs, what happens to its particles on average?	1. They move faster Heating increases the average kinetic energy of particles, so their motion increases.
6	Recall If a sample is cooled and no chemical reaction occurs, what happens to its particles on average?	2. They move slower Cooling decreases average kinetic energy, so particle motion decreases.
7	Recall When a substance changes state but does not react chemically, what stays the same?	1. The identity of its molecules A physical state change can change motion, spacing, and arrangement without changing molecular identity.
8	Recall How do particles in a solid usually move?	1. They vibrate around fixed positions Particles in a solid still move, but mainly by vibrating around fixed positions.
9	Recall Which description best matches particles in a liquid?	1. They are close together and can move past one another Liquid particles remain close together but can move or slide past one another.
10	Recall Which description best matches particles in a gas?	1. They are far apart and move freely Gas particles are much farther apart than in liquids or solids and move freely through the container.
11	Concept Water freezes from liquid to solid with no chemical reaction. Which statement is correct?	2. H₂O molecules remain H₂O but their motion and arrangement change Freezing is a physical change. The water molecules remain H ₂ O while particle motion and arrangement change.
12	Concept A liquid is heated until it becomes a gas, with no chemical reaction. Which model change is best?	1. Same molecules moving faster and much farther apart During vaporization without reaction, molecular identity stays the same while particles move more rapidly and spread far apart as a gas.
13	Concept The model $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ is balanced because it has which atom counts on both sides?	2. 4 H and 2 O Two H ₂ molecules contain 4 H atoms and one O ₂ molecule contains 2 O atoms; two H ₂ O molecules contain the same totals.
14	Concept A student draws 3 red atoms before a reaction but only 2 red atoms after. What is wrong with the model?	1. It violates conservation of atoms A reaction model must conserve each type of atom; one red atom cannot simply disappear.
15	Concept Which change in a particle model is the strongest sign of a chemical change?	2. Atoms become bonded to different partners New bonding combinations indicate that new substances have formed.

#	Question	Correct answer and rationale
16	Concept A sealed sample is warmed from 20 C to 50 C and no reaction occurs. What should a particle model show?	1. Same molecules with greater average motion A temperature increase alone does not require a composition change; the model should show greater particle motion with the same molecules.
17	Concept A model shows identical AB molecules before and after heating. Which additional change best represents the higher temperature?	1. Longer motion arrows Longer motion arrows can represent greater average particle speed while keeping the molecule AB unchanged.
18	Concept Which comparison correctly distinguishes a chemical change from a physical change?	1. Chemical: atoms rearrange into new substances; Physical: particle motion or arrangement changes without new substances Chemical changes create new substances through atom rearrangement; physical changes can alter state, spacing, or motion without changing identity.
19	Concept Why should the total mass stay the same before and after a reaction in a closed container?	1. Atoms are conserved and only rearranged In a closed system, atoms are not added or removed; rearranging the same atoms preserves total mass.
20	Concept Before a change, every particle is an AB molecule. After the change, the model shows A2 molecules and B2 molecules. What does this show?	1. A chemical change because bonding partners changed Changing from AB particles to A2 and B2 particles means bonds were broken and new bonds formed, which is a chemical change.
21	Application How many total atoms are in 3 CO2 molecules?	3. 9 Each CO2 molecule has 3 atoms total (1 C + 2 O). Three molecules contain 9 atoms.
22	Application How many hydrogen atoms are in 4 H2O molecules?	3. 8 Each H2O molecule has 2 H atoms. Four molecules contain $4 \times 2 = 8$ H atoms.
23	Application Before a reaction there are 2 AB2 molecules. Which atom count must be present after the reaction?	1. 2 A atoms and 4 B atoms Two AB2 molecules contain 2 A atoms and 4 B atoms, and those counts must be conserved.
24	Application Before: one A2 molecule and one B2 molecule. Which after-model conserves all atoms?	1. Two AB molecules $A_2 + B_2$ contains 2 A atoms and 2 B atoms. Two AB molecules contain exactly the same atoms.
25	Application Before: 2 A2 molecules and 1 B2 molecule. Which after-model conserves all atoms?	1. 2 A2B molecules The reactants contain 4 A atoms and 2 B atoms. Two A2B molecules contain 4 A and 2 B.
26	Application A particle model starts with 3 XY molecules. How many total atoms must be shown after a chemical change?	2. 6 Each XY molecule has 2 atoms, so 3 molecules contain 6 atoms total. A chemical change must conserve those atoms.
27	Application A gas is warmed from 20 C to 60 C without reacting. Which before-and-after model is best?	1. Same number and type of molecules with faster motion after Heating without reaction increases average motion but does not create matter or change atom types.
28	Application A liquid is cooled without reacting. Which change should a particle model show?	1. Lower average particle motion with the same molecular composition Cooling lowers average particle motion while the molecules remain the same if no reaction occurs.
29	Application Before: 2 AB molecules and 1 C2 molecule. Which after-model conserves every atom?	1. 2 AC molecules and 1 B2 molecule Before there are 2 A, 2 B, and 2 C atoms. Two AC molecules plus one B2 molecule contains exactly 2 A, 2 B, and 2 C.
30	Application A student adds an extra oxygen atom to the product side of a particle model. What is the best correction?	1. Make the number of each atom type match before and after A valid reaction model must show the same number of each atom type before and after the change.